

# Modem Harika

3rd Year Undergraduate  
Department of Mechanical Engineering

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## Academic Qualifications

| Year           | Degree/Certificate | Institute                              | CPI/%  |
|----------------|--------------------|----------------------------------------|--------|
| 2023 - Present | B.Tech             | Indian Institute of Technology, Kanpur | 6.0/10 |
| 2023           | TSBIE(XII)         | Narayana Junior College, Kothapet      | 97.9%  |
| 2021           | TS SSC(X)          | Sri Chaitanya School,L.B.Nagar         | 10/10  |

## Scholastic Achievements

- Secured **All India Rank 16316** in **JEE Advanced 2023** among the 1.89 Lakh shortlisted candidates.
- Secured **All India Rank 12671** in **JEE Mains 2023** among 1.36 million candidates.
- Secured **State Rank 823** in Telangana state engineering entrance test **TS EAMCET 2023** among 1.95 Lakh candidates.

## Key Projects

- **Applications of Large Language Models** (Dec'24 - Jan'25)  
Association of Mechanical Engineers (AME) Winter Project
  - Developed a system that translates text from images into English using OCR and LLMs, bridging language barriers.
  - Designed a tool to summarize key themes from user search queries, making research more efficient.
  - Implemented a question-answering module to provide precise, context aware responses to user queries.
  - Developed and tuned encoder-decoder models using LSTMs/RNNs for sequence prediction tasks.
  - Summarized large PDF documents using LangChain tools like MapReduce and RefineChain.
- **Screw Jack Assembly Drawing** (Jun'25)  
Self Project, Autocad
  - Designed a complete 2D and 3D screw jack assembly including top plate, lead screw, collar, nut, handle, and base.
  - Created front, top and sectional view with correct dimensioning, hatching, and layer management following engineering drawing standards.
  - Practiced real-world CAD workflows by integrating design thinking, visualization and manufacturing concepts.
- **Vibration Analysis & PID Control of a Washing Machine (ME354)** (Feb'26 - Mar'26)  
*Course Project, Mechanical Engineering*
  - Developed an SDOF mathematical model and derived transfer functions using Laplace-domain analysis to study vibration response and system stability.
  - Designed and tuned a PID controller achieving **94% reduction** in vibration amplitude, significantly improving stability and settling time.
  - Built and validated a Python-based simulation framework to compare controlled vs. uncontrolled system performance against theoretical predictions.

## Technical skills

- **Programming Languages:** C, C++, Python
- **Software:** Autocad, MS Excel
- **Libraries & ML Frameworks :** Pandas, NumPy, Tensorflow, Keras

## Positions of Responsibility

- **Senior Executive (Public Relations)** , Udghosh, IIT Kanpur (Oct'24)
  - Led guest outreach and sucessfully invited high-profile personalities to the fest.
  - Coordinated with guests for scheduling, invitations, and smooth conduct of sessions.
  - Helped in writing and sending formal invites and managing guest communications. Worked with a team to ensure all guest-related activities ran smoothly during the fest.
- **Organizer ( Show Management)**, Udghosh, IIT Kanpur (Sept'25 - Oct'25)
  - Coordinated effectively with a team of heads, senior executives to ensure the seamless execution of the Festival.
  - Designed creative layouts and activity modules, enhancing participant engagement and overall visual presentation.
  - Supervised setup and decoration using handmade craft models, improving event aesthetics and organization quality.
  - Developed and implemented strategies for the decoration of event related areas utilising the innovative ideas of various members.

Relevant Courses

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fundamentals of Computing<br>Machine Learning, Coursera (online)<br>Thermodynamics<br>Fluid Mechanics<br>Engineering Graphics<br>Nature and Properties of Materials | Linear algebra and Differential Equations<br>Robotics<br>Mechanics of Solids<br>Dynamics<br>Theory of Machines and Mechanism<br>Primary and Secondary Manufacturing Process |
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